

Memory-Centric Relational Perception: Frozen Results Assembly

Internal manuscript assembly preview

Counterfactual identity-memory evaluation. We hold the observation and semantic query fixed while switching the supplied miner identity memory, which changes the intended helmet target. Table 1 reports the primary 50-group documented-protocol-disjoint confirmation. CMA achieves 92% CMSA on clean observations and 58% under `target15_b`, compared with 4% and 2% for pinned SegLLM. Under degradation, Memory Fidelity is 84% versus 46%, and Identity Error Rate is 3% versus 38%. These results support stronger supplied-memory use on this evaluated set. Figure 1 illustrates the controlled identity switch and a retained degradation failure using previously frozen cases.

Robustness under compound degradation. CMA attains 67.53% mIoU under the fixed compound stressor, compared with 36.44% for SegLLM, a paired gap of 31.08 percentage points on identical groups. The supported claim is higher absolute degraded performance and retained identity fidelity: CMA still fails strict degraded CMSA on 21 of 50 groups. The overlap-affected historical and development replications are reported separately in Table 2 and are not pooled with the primary set.

Comparison scope. SegLLM is a pinned released historical-memory comparator receiving the same observations and supplied miner identities through its native interface. The comparison is system-level: training exposure and domain adaptation differ, and SegLLM’s lower clean performance limits architecture-only attribution. The counterfactual intervention measures identity-dependent behavior under common inputs; it does not isolate the contribution of an individual CMA component.

Limitations. The primary set was selected before inference and is disjoint by source-image SHA256 from reconstructed documented final-stage training and historical evaluation registries. Exact run-bound w15 and ancestor/pretraining exposure remain unknown, so this is not an exact training-unseen test. Memories are externally supplied and targets are reconstructed pseudo labels; 23 groups retain historical review-level episode QC and 27 combine accepted-high pairs. Structural validity does not replace independent annotation. CMA’s clean-to-degraded mIoU decline is 20.75 points, versus 6.26 for SegLLM; we do not claim lower degradation sensitivity. The evidence does not establish autonomous memory writing, persistence or repair, broad corruption generalization,

or safety guarantees. Earlier Layer-2 verifier/controller attempts did not meet their stated requirements and remain negative evidence, not a successful agent contribution.

Table 1: Primary Cycle025 DOCUMENTED_PROTOCOL_DISJOINT_CONFIRMATION: 50 groups / 100 identity references per condition, with supplied identity memories and reconstructed pseudo targets. The comparison is system-level; exact run-bound w15 and ancestor exposure is unknown. Counts and rates are shown for CMSA, Memory Fidelity and IER; margins are unscaled. The set is disjoint from documented training/evaluation image-byte registries, not proven exact training-unseen.

Method	Condition	mIoU (%)	CMSA	Fidelity	IER	Mean margin
CMA base-w15	clean	88.28	46/50 (92%)	97/100 (97%)	1/100 (1%)	0.873944
CMA base-w15	target15_b	67.53	29/50 (58%)	84/100 (84%)	3/100 (3%)	0.646666
SegLLM pinned	clean	42.71	2/50 (4%)	49/100 (49%)	42/100 (42%)	0.049034
SegLLM pinned	target15_b	36.44	1/50 (2%)	46/100 (46%)	38/100 (38%)	0.031818

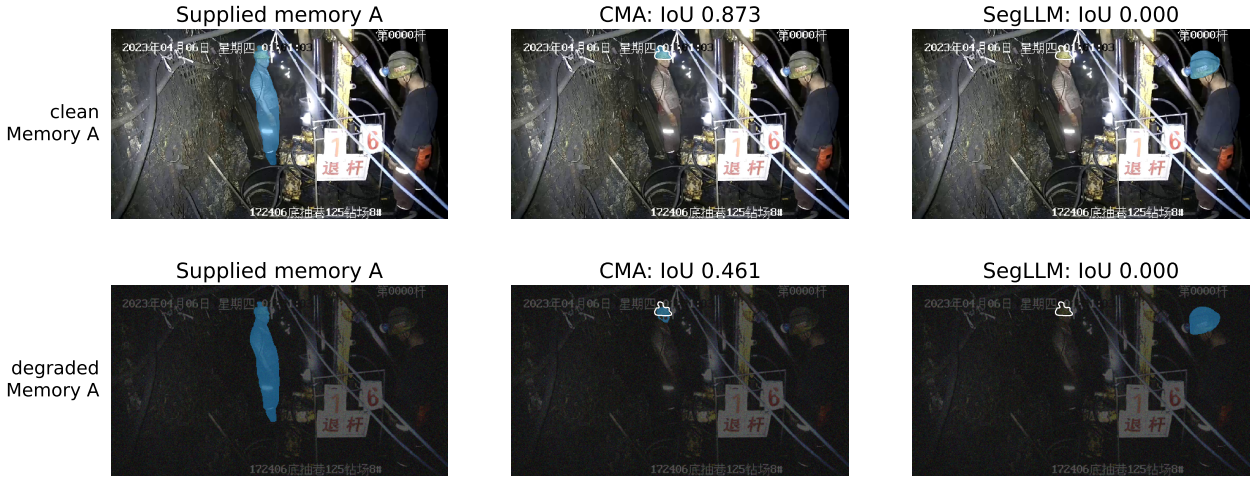
(a) Identity switch: same degraded observation, memory A vs B

cf_778571dc8020b5df



(b) Degradation limitation: fixed memory A, clean vs degraded

acceptedpair_b914b3d4a14a9083



White outline: saved pseudo target. Blue: identity A; orange: identity B.
Full frames; display-only mask overlays. IoU copied from frozen records.

Figure 1: Supplied identity-memory use and its degradation limitation. (a) The same degraded observation/query with memories A and B in the frozen success case. (b) Fixed memory A on clean and degraded observations in the frozen limitation case; memory B remains in the supplementary bank. Columns show memory, CMA and SegLLM. Blue/orange identify A/B; white contours are saved helmet pseudo-targets. IoUs are copied from frozen records. Full frames use display-only overlays without brightness or mask changes. The success case retains review-level episode QC; the limitation case combines accepted-high pairs. No new case selection or inference was performed.

Table 2: Separate overlap-affected replications: each block contains 50 groups and 100 identity references per condition. The reconstructed training protocol overlaps 37/50 Cycle022 images and 50/50 Cycle018 images; their source holdout was historically evaluated. Cycle018 also served development. These are not independent training-unseen tests. Supplied memories, reconstructed pseudo targets and system-level comparison limitations apply. No pooling is performed.

Method	Condition	mIoU (%)	CMSA	Fidelity	IER	Mean margin
<i>Cycle022: historical replication; 37/50 documented overlap</i>						
CMA base-w15	clean	90.87	47/50 (94%)	98/100 (98%)	2/100 (2%)	0.892993
CMA base-w15	target15_b	68.02	30/50 (60%)	87/100 (87%)	2/100 (2%)	0.657803
SegLLM pinned	clean	37.29	3/50 (6%)	43/100 (43%)	38/100 (38%)	0.041568
SegLLM pinned	target15_b	27.52	1/50 (2%)	38/100 (38%)	30/100 (30%)	0.016394
<i>Cycle018: development replication; 50/50 documented overlap</i>						
CMA base-w15	clean	91.97	47/50 (94%)	99/100 (99%)	0/100 (0%)	0.919718
CMA base-w15	target15_b	69.11	29/50 (58%)	89/100 (89%)	3/100 (3%)	0.662931
SegLLM pinned	clean	41.57	2/50 (4%)	47/100 (47%)	41/100 (41%)	0.047717
SegLLM pinned	target15_b	33.14	0/50 (0%)	42/100 (42%)	39/100 (39%)	-0.027858